Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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ABSTRACT

Stablecoins are designed to trade near one dollar, but stress events repeatedly create visible deviations across venues, quote currencies, and settlement rails. These deviations often appear to be arbitrage opportunities. In practice, however, a quoted price gap is not the same as an executable trade: a trader must clear order-book depth, absorb VWAP slippage, pay fees, and bear settlement risk before any apparent spread becomes profit.

We introduce Stablecoin StressBench, an execution-aware benchmark for evaluating stablecoin dislocation models. The benchmark separates three layers: optical price dislocations, executable arbitrage opportunities, and ex-ante predictable executable opportunities. During the March 2023 USDC/SVB stress window, 34.3% of 1-minute windows exceed a 10 bps max cross-quote basis threshold on price alone, while only 2.88% remain profitable at $10K notional after VWAP book-walking and trading costs. A hindsight oracle earns positive net bps, confirming that profitable windows exist. On executable arbitrage tasks every active model loses money out of sample; on the basis classification task a LightGBM classifier achieves a marginal +17.2 bps — 145 bps below the oracle, and not economically deployable — while the executable-arbitrage oracle gap exceeds 260 bps.

The result reframes stablecoin arbitrage prediction as an execution-barrier problem rather than a price-signal detection problem. Stablecoin StressBench contributes a historical stress-event catalogue, execution-aware labels, economic evaluation metrics, and a model ladder for future work on crypto market microstructure, optimal execution, and financial benchmark construction.

KEYWORDS

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

ACM Reference Format:
Anonymous Author(s). 2026. Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations. In Proceedings of ACM International Conference on AI in Finance (ICAIF '26). ACM, New York, NY, USA, 8 pages.

1 INTRODUCTION

Stablecoins are designed to make a simple promise: one token should trade close to one dollar. This promise makes stablecoins central to crypto-market settlement, quoting, collateral, and liquidity transfer. When confidence in that promise weakens, prices fragment across venues, quote currencies, and liquidity pools.

Price fragmentation often appears to create arbitrage. During a stress event, one route may imply that USDC is cheap while another implies that USDT or USD is expensive. A trader looking only at quoted prices may conclude that buying through one route and selling through another should generate mechanical profit.

That conclusion is incomplete. A quoted price gap is not a trade. To realize arbitrage, a trader must clear available order-book depth, absorb VWAP slippage, pay taker fees, and bear settlement delay. These frictions are small individually but large jointly. As a result, many stablecoin dislocations are optical rather than executable.

This paper introduces Stablecoin StressBench, an execution-aware benchmark for stablecoin dislocations. The benchmark separates three layers: optical price dislocations, executable arbitrage opportunities, and ex-ante predictable executable opportunities. The distinction matters because a model can classify price dislocations accurately while still losing money if the selected windows do not survive execution.

We study this distinction during the March 2023 USDC/SVB stress event and in a broader historical catalogue of stablecoin stress episodes. In the primary Tier-A test window, 34.3% of minutes exceed a 10 bps max cross-quote basis threshold, but only 2.88% remain profitable at $10K notional after VWAP execution and costs. A hindsight oracle confirms that profitable windows exist. On executable arbitrage tasks every active model loses money out of sample; on the basis task the best classifier (LightGBM) achieves only +17.2 bps — 145 bps short of the oracle. The result reframes stablecoin arbitrage prediction as an execution-barrier problem: the hard problem is not detecting that a stablecoin price moved, but identifying which dislocations survive execution.

Contributions. We make four contributions.

(1) Execution-aware stablecoin benchmark. Per-minute labels distinguish quoted price dislocations from executable arbitrage opportunities using order-book depth, VWAP book-walking, fees, notional size, and settlement penalties.
(2) Historical stress-event taxonomy. We catalogue 18 canonical pre-2024 stablecoin stress events across seven mechanism classes and classify each by data availability. A cross-mechanism check (Terra/LUNA validation split) provides a directional consistency check on the price-to-execution gap under an independent mechanism (algorithmic/reflexive): 5.9× ratio vs. 12× for the Tier-A test.
(3) Execution gap and oracle gap: evidence. In the USDC/SVB Tier-A test event, 34.3% of minutes exceed a 10 bps max cross-quote basis threshold, but only 2.88%

ICAIF '26, November 14–17, 2026, Milan, Italy
2026.

remain executable at $10K. A hindsight oracle earns positive bps on every task. On executable arbitrage tasks every active model produces negative net bps; on the basis task LightGBM achieves a marginal +17.2 bps, still 145 bps below the oracle.

(4) Literature-motivated model ladder. We evaluate economic anchors, naive price rules, time-series baselines, regularized linear models, tree ensembles, direct expected-profit regression, meta-labeling, and regime detectors. This ladder establishes reproducible baseline failures for future models attempting to close the oracle gap.

This paper fits ICAIF as financial benchmark construction for blockchain and cryptocurrency markets, combining market microstructure, trading and execution, validation and calibration of financial models, and economic evaluation of ML systems.

2 RELATED WORK

Stablecoin stress and depegs. The USDC/SVB episode (March 2023) produced well-documented price fragmentation across venues; we address the ex-ante predictability question that event studies leave open. Uhlig [1] analyses Terra/LUNA collapse mechanics; we use May 2022 as our validation split event. Griffin and Shams [2] study Tether price manipulation; our contribution is execution microstructure, not flow manipulation.

Limits to arbitrage and crypto market fragmentation. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish the general theory. Makarov and Schoar [8] show persistent cross-venue crypto price gaps — confirming that gaps exist but are not automatically executable profits. Hautsch et al. [3] show settlement latency and transfer friction are first-order arbitrage costs; we extend this to stablecoin cross-quote routes. The relative-value threshold-rule tradition [13] motivates our price-basis rules; we show such rules overfire once depth and fees are imposed.

Market microstructure and optimal execution. Glosten and Milgrom [7] establish the theoretical basis for adverse selection during stress. Cont and Kukanov [6] develop optimal routing across fragmented limit-order books; the key insight is that the relevant object is the executable VWAP price, not the quoted price. Stablecoin cross-quote arbitrage is structurally a triangular arbitrage problem: direct and indirect exchange-rate paths should agree in frictionless markets, but VWAP book-walking reveals when they do not survive costs [9].

Financial ML benchmarks and model validation. López de Prado [10] introduces meta-labeling to separate primary signal generation from execution filtering — directly applicable where price signals are plentiful but most do not survive costs. Adams and MacKay [17] introduce Bayesian online changepoint detection (BOCPD); Cintra and Holloway [11] apply it to Curve pool reserves and detect the May 2022 Terra depeg with a 12-hour lead. Their target is depeg risk; ours is executable arbitrage — related but distinct, as we find BOCPD achieves only AUROC 0.23 on CEX cross-quote data. No prior benchmark provides execution-aware stablecoin arbitrage labels with documented route-depth provenance and a systematic price-to-execution gap analysis.

Table 1: Route-leg depth provenance per benchmark window.

Leg (instrument)	SVB test Mar 2023	2024 windows
Buy — BTC-USDC	kline-proxy†	real L2 (futures)
Cross — USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)
Sell — BTC-USDT	real L2 (futures)	real L2 (futures)
Ref — BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual not listed on Binance until Jan 2024. All 2022 windows: kline-proxy on all legs (Binance Vision archive predates 2023).

Table 2: Dataset coverage by split.

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%
Total		56,134	

Pos. = `net_profit_bps_q10000` > 10 bps.

3 BENCHMARK DESIGN

3.1 Data Sources and Instruments

The gold-layer dataset contains 56,134 1-minute bars across six windows (three calm-control training windows, one validation stress event, one test stress event, and its recovery window) (Table 2). Price and reference features are derived from the Binance Vision public archive (klines, aggTrades) and Coinbase REST candles. Net-profit labels are computed on routes with confirmed order-book depth coverage; the cross-quote route uses Binance BTC-USDC and BTC-USDT depth alongside Coinbase BTC-USD as the reference leg.

Depth provenance and route completeness. The pipeline tags every depth record with a provenance field: `binance:futures_bookdepth` for real futures bookDepth snapshots, or `binance:klines/coinbase:candles` for kline-synthesised approximations. Table 1 summarises per-leg provenance for the primary execution label. The sell leg (BTC-USDT) uses real L2 futures bookDepth for all 2023 and 2024 benchmark windows (Binance Vision public archive). The cross leg (USDC-USDT) uses real L2 futures bookDepth from 2023-03-12 onward, covering 8 of the 10 USDC/SVB test days and all 2024 windows. The buy leg (BTC-USDC) uses kline-synthesised depth for the USDC/SVB test window (the BTC-USDC perpetual did not exist on Binance before 2024) and real L2 futures bookDepth for the 2024 calm-control windows. The net-profit labels reported in this paper are therefore execution-labelled route estimates with documented real/proxy depth provenance, not fully route-complete real-L2 replay on every leg for all windows. Full route-complete coverage for 2022 events and the BTCUSDC buy leg in 2023 requires a paid Tardis archive subscription.

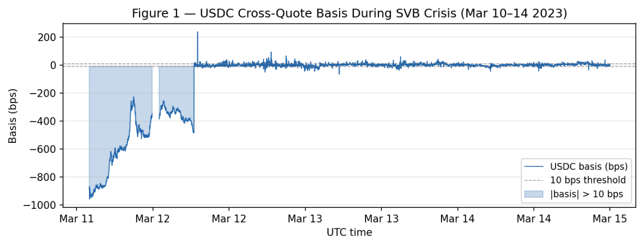

Figure 1: Cross-quote basis (bps) during the SVB stress window (March 2023), exceeding 100 bps for sustained periods. With real USDCUSDT futures bookDepth, the USDC-specific basis exceeds 10 bps in 12.5% of test minutes (capturing the actual SVB-period USDC de-peg); the USDT route exceeds 10 bps in 27.4% of minutes; the max-abs basis (maximum of both routes) exceeds 10 bps in 34.3%.

Table 3: Data tier and permitted claims.

Tier	Data	Permitted claims
A (N=2)	Execution labels with documented route provenance	Exec. gap, oracle gap, P&L, model eval (provenance caveat)
B (N=11)	OHLCV / pool data	Peg magnitude (est.), liquidity context
C (N=5)	Partial / DEX-only	Mechanism taxonomy only

3.2 Event-Based Split Design

Splits are defined by event identity, not by time alone (Figure 1). The train split contains three calm control windows (no stress event) that bracket the test period; the validation split uses the Terra/LUNA collapse (May 2022), a qualitatively similar but independent stress event; the test split uses the USDC/SVB de-peg (March 10–15, 2023).

This design has two guarantees: (i) no model trained on the train split has seen stress dynamics; (ii) threshold calibration on validation uses an independent stress event, not a held-out portion of the same episode. No lookahead: labels are constructed by `join_asof` at $t + h$ and formally verified. The Terra/LUNA validation split is not used to claim a second Tier-A execution-labelled result; its role is calibration, exposing the thresholding procedure to independent stress dynamics before final evaluation on USDC/SVB.

3.3 Historical Stress Taxonomy and Data Tiering

Stablecoin stress is not a single phenomenon. We catalogue 18 stress events from 2020–2023 spanning seven distinct failure mechanisms: algorithmic/reflexive (FEI, IRON/TITAN, MIM, UST, USDD), fiat-reserve bank shock (USDC/SVB), regulatory/issuer winddown (BUSD, Binance conversion), exchange credit/liquidity (Celsius/3AC, HUSD, FTX [12]), DeFi pool imbalance (Curve/UST, USDT/Curve, USDC/DAI secondary), collateral liquidation (DAI Black Thursday), and RWA/niche stablecoin (Acala aUSD, USDR).

Events are classified by data availability tier, which governs what claims this paper may make (Table 3).

Table 4: Stress-event roles and permitted claims in Stablecoin StressBench.

Event	Mechanism	Tier	Permitted use
Terra/UST 2022	Alg./reflexive	Val.	Threshold calibration; cross-mechanism check
USDC/SVB 2023	Fiat-reserve shock	A	Main exec. test; all claims
USDC recovery	Fiat-reserve recov.	A	Comparator only
USDT/Curve 2023	DeFi pool imbalance	B	Price/liq. context (est.)
FTX 2022	Exchange credit	B	Market-stress context (est.)
BUSD 2023	Regulatory winddown	B	Issuer/reg. context (est.)
IRON/TITAN 2021	Alg./reflexive	C	Mechanism taxonomy only

Tier-A execution-labelled route estimates are available for the USDC/SVB stress window and its immediate post-stress recovery window. The price-to-execution gap, oracle net bps, and model evaluation in this paper are anchored to the USDC/SVB stress test; the recovery window serves as a Tier-A comparator, not an independent stress mechanism. Price-grade claims for Tier-B events are labelled as estimates (est.) and excluded from any arbitrage-profitability argument.

L2 data availability is not equivalent to Tier-A route coverage. A stablecoin arbitrage label requires route-complete depth: synchronized buy- and sell-leg order books, sufficient depth at the tested notional, and non-synthetic provenance for the relevant instruments and venues. For Tier-B and Tier-C events — FTX, Celsius/3AC, USDT/Curve, BUSD, and others — no order-book depth data is available in the repository. Those events support price and taxonomy claims only (see `docs/execution_route_coverage.md`).

3.4 Stress-Event Roles in the Benchmark

Tables 4–5 clarify how the historical catalogue is used. Each mechanism class produces a different execution friction (Table 5): algorithmic collapses destabilise the reference price itself; fiat-reserve shocks widen spreads as depth withdraws; exchange-credit events segment venues; DeFi imbalances shift slippage to AMM curves. Only Tier-A events support execution-aware P&L claims. The catalogue is a canonical stress taxonomy, not a complete census: it covers the major pre-2024 mechanism classes that are useful for benchmark design; post-2023 regulatory and product-design changes (e.g. federal stablecoin regulation and synthetic-dollar products) are outside the empirical scope and left for future work.

Non-claims. Stablecoin StressBench does not claim to reconstruct every historical stablecoin arbitrage route, nor that all depeg episodes contain executable arbitrage. Only the USDC/SVB test window is used for the main price-to-execution and oracle-gap claims. Terra/UST provides

Table 5: Mechanism classes and expected execution frictions.

Mechanism	Stress channel	Execution friction
Alg./reflexive	Redemption spiral	Unstable ref. price; thin book as peg breaks
Fiat-reserve shock	Reserve uncertainty	Depth withdrawal; spread widening
Exchange credit	Venue insolvency	Venue segmentation; bilateral credit risk
DeFi pool imbal.	LP withdrawal	AMM slippage; pool concentration
Regulatory w/d	Issuer migration	Fragmented order flow; supply shift
Collateral liq.	Oracle dependency	Cascading fills; stale marks
RWA/niche	Illiquid collateral	Thin market; settlement uncertainty

validation-split calibration and a cross-mechanism consistency check; other events provide mechanism taxonomy and external validity. Table 4 documents permitted uses per event.

Mechanism mini-cases. Each mechanism class produces a distinct execution friction (Table 5). Algorithmic collapses (Terra/UST, IRON/TITAN) destabilise the reference price itself, disabling the route. Fiat-reserve shocks (USDC/SVB) widen spreads as market depth withdraws; USDC/SVB is the primary execution-labelled test because it combines a large dislocation with CEX route-depth provenance. Exchange-credit events (FTX, Celsius/3AC) cause venue segmentation; no public archive holds depth data for those windows. DeFi pool imbalance (USDT/Curve) manifests first in AMM reserves, not CEX cross-quote prices, and is context-only. Terra/UST serves as the validation split for threshold calibration; it is not used to claim a second Tier-A result.

3.5 Feature Sets

We define four nested feature sets from narrowest to broadest:

`price_only`	5 cross-quote basis columns
`price_plus_book`	+7 microstructure cols (spread, depth, imbalance)
`price_book_frag`	+3 cross-venue fragmentation cols
`price_book_settle`	+7 on-chain settlement proxies

The nesting structure isolates the marginal value of each information layer.

Table 6: Three layers of stablecoin dislocation (USDC/SVB test split, \$10K notional).

Layer	What it measures	SVB%
1. Optical	Quoted basis >10 bps on prices	34.3%
2. Executable	Net profit >10 bps after costs	2.88%
3. Predictable	Model predicts window ex ante	None

Layer 3: on executable arbitrage tasks no active model achieves positive net bps; on the basis task LightGBM achieves +17.2 bps (145 bps below oracle). `NoTrade` is the floor.

4 EXECUTION-AWARE LABELS

The benchmark measures three layers of stablecoin dislocation (Table 6). Understanding these layers clarifies why standard price-signal metrics are insufficient: a model can do well on Layer 1 while failing economically at Layer 2.

The gap between Layer 1 and Layer 2 is the price-to-execution gap; the gap between Layer 2 and Layer 3 is the oracle gap. Both are central measurements of this benchmark.

4.1 Cross-Quote Basis

The cross-quote USDC basis at time t is:

$$b_{\text{USDC}}(t) = 10{,}000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad [\text{bps}]$$

where $P_{\text{BTC/USDC}}$ is the effective BTC price obtained via USDC and $P_{\text{BTC/USD}}$ is the direct BTC-USD price. We also compute the max-absolute basis $b_{\max}(t) = \max(|b_{\text{USDC}}|, |b_{\text{USDT}}|)$ as a broader stress detector.

Basis labels for horizon $h \in \{1, 5, 15, 60\}$m and threshold τ:

$$y_{\text{basis}}(t) = \mathbf{1}[\, |b_{\text{USDC}}(t+h)| > \tau \,]$$

Primary classification task: `basis_usdc_1m_gt10bps`.

4.2 VWAP Execution Walk

For each minute window and notional size q (\$10K, \$50K, \$100K, \$500K), we simulate a round-trip execution:

(1) Walk the bid/ask ladders for routes with confirmed depth provenance to fill q notional, recording the VWAP price at each level consumed.
(2) Gross spread: $\text{gross}(q) = \text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}$.
(3) Deduct taker fees on both legs and a settlement delay penalty.

4.3 Net Executable Arbitrage Labels

$$\text{net}(q,t) = \text{gross}(q,t) - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \tag{1}$$

$$y_{\text{arb}}(q,h,t) = \mathbf{1}\left[\max_{s\in[t,t+h]} \text{net}(q,s) > c\right] \tag{2}$$

where $f_{\text{buy}}, f_{\text{sell}}$ are venue taker fees, δ_{settle} is a settlement penalty proxy, and $c \geq 0$ is a minimum-profit threshold. The primary paper setting uses $c = 10$ bps, matching the price-basis threshold; sensitivity to $c \in \{0, 5, 25\}$ bps is shown in Table 7. Primary economic task: `executable_arb_q10000_5m`. The forward-max in Eq. (2) asks whether any profitable window exists in the next h minutes, matching the constraint that a trader may execute at any point in the horizon.

4.4 Oracle Upper Bound

Let $\mathcal{F}_t$ denote all prices, depth, and features available at minute t. The oracle uses the realized future $[t, t+h]$ — it is not $\mathcal{F}_t$-measurable, hence not deployable. It trades every minute where $\mathrm{net}(q, t) > c$ in hindsight, setting the theoretical ceiling and defining the oracle gap that any $\mathcal{F}_t$-measurable model must close.

5 MODELS AND EVALUATION

5.1 Model Ladder

The benchmark frames stablecoin arbitrage prediction as three nested problems: (1) cross-quote parity — do prices imply a basis? (2) microstructure — does the basis survive VWAP depth-walk and costs? (3) statistical learning — can features at t predict whether future net profit is positive? The model ladder addresses each layer in turn, ranging from economic anchors to price rules, time-series baselines, interpretable linear models [14], nonlinear ensembles [15, 16], expected-profit regression, meta-labeling, and regime detectors — establishing whether stronger models can close the oracle gap.

Economic anchors. `NoTrade` (the economic floor: 0 net bps) and `Oracle` (hindsight ceiling).

Rule baselines. `PriceBasis10bps` (trade when $|b_{\mathrm{USDC}}| > 10$ bps), `PriceBasis25bps`, and `GrossArbThreshold` (gross spread $> c$). These test whether threshold rules on price data suffice; following [3], execution frictions are expected to make raw price rules uneconomic.

Statistical baselines. `LastValue` (AR0), `RollingMean`, `AR1` — testing whether basis autocorrelation alone identifies executable stress. Each baseline uses an explicitly named lag column rather than assuming column ordering.

ML classifiers. Logistic (L2), Lasso (L1), Random Forest, XGBoost, and LightGBM — each trained on the four nested feature sets. Tree ensembles are included for their strong tabular performance in financial time-series settings [8].

Expected-net-profit regressor. `ExpectedNetProfitRegressor` directly predicts $\mathrm{net}(q, t)$ and trades when $\hat{y} > \theta^*$. If this also fails, the oracle gap is structural, not a label-mismatch artefact.

Meta-labeling filter. Motivated by [10]: the primary signal is $|b_{\mathrm{USDC}}| > 10$ bps; a secondary LightGBM meta-model learns $\hat{y}_{\mathrm{meta}} = \Pr[\mathrm{net}(q, t) > c \mid \text{primary fires}]$ and filters predicted false positives.

Regime detectors. EWMA z-score, CUSUM, and BOCPD [17, 11] identify stress regimes from the basis time series as two-stage precursors: detect stress first, then apply execution filters only during detected regimes.

Table 7: Price-to-execution gap (test split, $10K notional, 1-minute same-window horizon).

Thresh.	$\mathrm{P}_{\max}$	$\mathrm{P}_{\mathrm{USDC}}$	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

5.2 Threshold Calibration

For probabilistic classifiers, the decision threshold is calibrated on the validation split:

$$\theta^* = \arg\max_{\theta \in [0.05, 0.95]} \textstyle\sum_{i: \hat{p}_i > \theta} \mathrm{net}(q, t_i) \quad \text{s.t.} \quad |\{i : \hat{p}_i > \theta\}| \geq 25$$

The ≥ 25 trade minimum prevents degenerate abstention strategies from winning on small-sample artefacts.

5.3 Evaluation Protocol

Primary (economic): `net_bps_captured` (mean net bps/trade), `hit_rate_above_cost`, `final_pnl_usd`, `oracle_capture_pct` (net bps ÷ oracle net bps).

Secondary (statistical): AUROC, AUPRC, Brier score, n_{trades}.

The `NoTrade` baseline (= 0 net bps) is the economic floor: a model that loses money is worse than abstaining entirely, regardless of its AUROC.

5.4 Assumptions and Claim Boundaries

Three conditions scope the quantitative claims. (A1) Execution-labelled, not route-complete: the BTC-USDC buy leg uses kline-synthesised depth for the SVB window (Table 1). (A2) Scope USDC/SVB Tier-A only: oracle-gap and model results are from the SVB test split (N=15,832); Terra/LUNA is a directional consistency check only. (A3) Parameters: c=10 bps, taker fee 4 bps/side, δ=5 bps; Tables 7–10 show sensitivity.

6 RESULTS

6.1 Price Gaps Are Common, Execution Is Rare

Table 7 and Figure 2 show the central finding. At a 10 bps threshold, 34.3% of test-split minutes show a max cross-quote basis exceeding the threshold on price alone; only 2.88% yield net profit above 10 bps after execution costs at $10K notional — a 12× price-to-execution ratio (block bootstrap 95 % CI: 8×–24×, 60-min blocks, B=10,000 resamples). With real USDCUSDT futures bookDepth, the USDC-specific basis fires in 12.5% of minutes (correctly capturing the SVB-period USDC discount); the USDT route contributes the majority of the max-abs signal (USDT >10 bps in 27.4% of minutes), with USDC dominating the max-abs signal in 34.3% of those minutes. The gap persists at every threshold level, widening sharply below 10 bps as price signals fire constantly but order-book depth is thin.

Tracing the pipeline stage by stage: of 15,832 total test minutes, 5,435 (34.3%) show a max-basis price dislocation above 10 bps; 456 of those (2.88%) yield net profit above

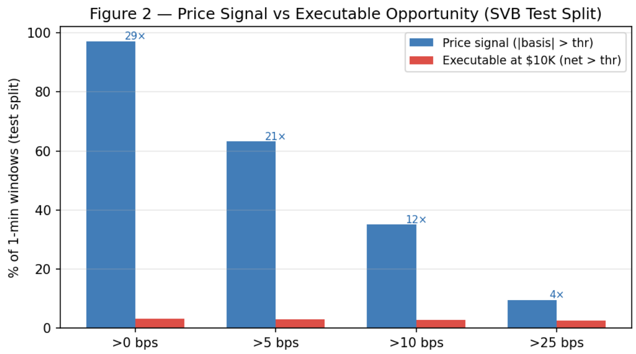

Figure 2: Price-to-execution gap by threshold. Price-signal frequency (dark) dwarfs executable frequency (light) at every level. The gap is largest at low thresholds, where price signals fire constantly but depth is thin.

Table 8: Oracle gap (test split). On executable tasks all active models are economically negative; *marks the one exception (basis task only).

Task	Oracle	Best model	Gap
`basis_usdc_1m_gt10bps`	+162.2 bps	+17.2 bps*	145 bps
`executable_arb_q10000_5m`	+225.0 bps	−41.1 bps	266 bps
`executable_arb_q50000_5m`	+147.1 bps	−113.0 bps	260 bps

*lgbm@all features (106 trades); marginal — 145 bps below oracle.
Best model: price_threshold_25bps@price_only (arb tasks).

10 bps at $10K/1m; the oracle trades 315 of those minutes profitably. Naive price rules trade far more often and lose money on nearly all trades; calibrated ML models trade less aggressively or abstain. Only one basis-task LightGBM specification is positive, and no executable-arbitrage model crosses zero.

6.2 Profitable Windows Exist, But Only in Hindsight

Table 8 gives the oracle gap for each primary task. The oracle earns +162.2 bps on the `basis_usdc_1m_gt10bps` task (315 trades). The best non-oracle model, LightGBM with full features, achieves +17.2 bps on this task — a 145 bps gap from the oracle. On the executable arbitrage task, the oracle earns +225.0 bps; the best rule (`price_threshold_25bps`) achieves −41.1 bps, a 266 bps gap. On the executable arbitrage tasks, every active non-oracle strategy produces negative net bps; the oracle gap is structurally large regardless of cost regime. The LightGBM result on the basis classification task is marginal (17.2 bps, 106 trades) and does not close the oracle gap.

6.3 The Model Ladder Fails Economically

Table 9 summarises the model ladder. Price-threshold rules overtrade severely (607–1969 trades, −136 to −347 bps): the rules fire on price dislocations with no executable depth. Validation-thresholded logistic takes 648 trades at −75.8 bps. LightGBM with full features achieves +17.2 bps (106 trades,

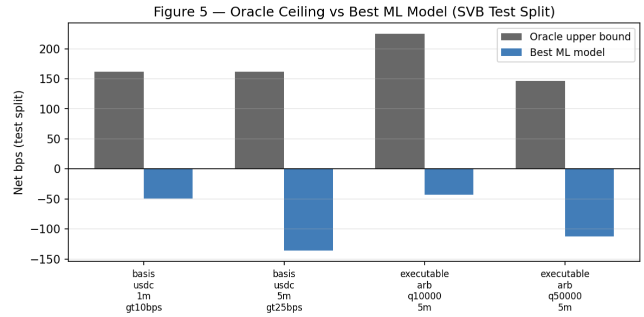

Figure 3: Oracle gap: grouped bars showing oracle net bps (gold) vs. best ML model (navy) for each task. The gap is 145 bps on the basis task (LightGBM +17.2 bps) and exceeds 260 bps on executable arbitrage tasks.

Table 9: Model ladder summary (task: **`basis_usdc_1m_gt10bps`**, test split). †Theoretical ceiling.

Model	Features	Net bps	Trades	AUROC
Oracle†	—	+162.2	315	—
`lgbm`	all	+17.2	106	0.717
`logistic`	all	−75.8	648	0.143
`gross_arb`	price_only	−136.4	607	0.531
`price_thr 10`	price_only	−270.0	1969	0.833
`price_thr 25`	price_only	−347.3	1025	0.746
`no_trade`	—	0.0	0	—
`ExpNetProfit`	price_only	−73.8	537	—

AUROC 0.72), the only non-oracle result above zero. However, this represents only 106 high-confidence predictions against an oracle that identifies 315 profitable windows earning +162.2 bps each; the 145 bps oracle gap remains. On the executable arbitrage tasks (which target net profit directly), no model crosses zero — the oracle gap there exceeds 260 bps.

The `ExpectedNetProfitRegressor` (target: $\mathrm{net}(q,t)$) achieves −73.8 bps — negative despite directly regressing on the economic criterion used for evaluation. This is the strongest evidence that the oracle gap is structural: even when the training objective directly matches the economic evaluation criterion, the model cannot identify profitable windows out of sample.

Why AUROC is insufficient here. `PriceBasis10bps` achieves AUROC 0.83 while losing −270 bps per trade (Table 9). AUROC measures ranking, not economic value: a model that correctly ranks the 2.88 % of profitable windows above the remaining 97.12 % still loses money if it fires too often on false positives. The relevant floor is `NoTrade` at 0 bps; beating a chance classifier is not sufficient.

6.4 Robustness and Diagnostics

Cost-regime robustness. The price-to-execution gap persists across all tested cost assumptions (Table 10). At base fees, 5.64% of windows are executable at $10K/5m/10 bps; high fees reduce this to 5.46%. Adding a 10 bps settlement penalty

Table 10: Price-to-execution ratio across cost regimes ($10K, 10 bps, 5-min horizon, test split).

Settlement	Fee regime	Exec. %	Ratio
0 bps	low (institutional)	6.11%	2.07×
0 bps	base	5.64%	2.24×
0 bps	high	5.46%	2.32×
5 bps	base	5.16%	2.45×
10 bps	base	4.88%	2.59×
10 bps	high (worst case)	4.81%	2.63×

Table 11: Regime detection (test split, stress = $|b_{\mathrm{USDC}}| > 10$ bps).

Detector	Acc.	Recall	FAR	AUROC
EWMA (span=20, thr=2.5)	0.869	0.010	0.007	0.624
CUSUM (k=0.5, h=5)	0.270	1.000	0.835	0.582
BOCPD (hz=0.01, thr=0.5)	0.860	0.022	0.019	0.229

FAR = false alarm rate.

gives 4.88%. At $50K notional the executable rate drops to 1.62%; at $500K, virtually no windows are executable (0.03%). The conclusion is qualitatively identical across all regimes.

Meta-labeling null result. The MetaLabelingFilter is trained on windows where the primary price signal fires ($|b_{\mathrm{USDC}}| > 10$ bps). In the train split only 53 such windows exist, and zero are net-profitable — consistent with the zero positive rate in calm-control training data (Table 2). The meta-model learns no positive class signal and takes 0 trades on the test split. This is an informative null: meta-labeling is architecturally sound but requires a training regime with both positive and negative meta-labels. It motivates using the Terra/LUNA validation split for meta-model training in future work.

Regime detectors. Table 11 shows the recall-precision trade-off. CUSUM achieves near-perfect recall (0.9995) at 83.5% false alarm rate; EWMA achieves 86.9% accuracy at 0.65% false alarm rate with very low recall (0.95%); BOCPD achieves only AUROC 0.229 on the CEX cross-quote series. No detector alone solves the execution identification problem; their value is as a pre-filter that reduces the search space.

False-positive diagnosis. Applying `PriceBasis10bps` to the test split produces 2,002 trades. False positives — 581 windows where basis > 10 bps but order-book depth is insufficient — account for 29% of trades at a mean net loss of −38.7 bps. This directly illustrates the execution-barrier problem: the rule fires on price dislocations where the book is too thin to fill the arbitrage leg.

6.5 Stress Recurs Across Mechanisms, But Execution Labels Are Scarce

The benchmark's main execution result rests on USDC/SVB. A natural question is whether the price-to-execution gap and oracle gap are specific to fiat-reserve bank shocks, or whether they generalise to other mechanisms.

Table 12: Cross-mechanism comparison: USDC/SVB vs. Terra/LUNA.

Metric	USDC/SVB (Tier-A test)	Terra/LUNA (Val. kline-proxy)
Minutes	15,832	11,526
Mechanism	Fiat-reserve	Alg./reflexive
Price-basis >10 bps (max)	34.3%	13.5%
Executable >10 bps ($10K)	2.88%	2.30%
Price-to-exec ratio	12×	5.9×
Oracle (hindsight, >10 bps)‡	+260 bps	+88 bps†
Route depth	2-of-3 real L2	kline-proxy

†Kline-proxy labels; directional comparison only — not Tier-A under the SVB route-provenance standard.

‡Oracle bps: executable_arb_q10000 task (see Table 8 for full task breakdown).

Cross-mechanism check: Terra/LUNA. The Terra/LUNA validation split provides the strongest available cross-mechanism evidence from within the benchmark. Table 12 compares key metrics across both events. The Terra/LUNA split (N=11,526 min, May 2022) uses kline-proxy route depth (Binance Vision 2022 predates the bookDepth archive); labels are kline-proxy and not Tier-A under the SVB standard. Nevertheless, the directional pattern is consistent: 13.46% of minutes show a price-basis above 10 bps, but only 2.30% survive execution at $10K notional — a 5.9× execution gap. The hindsight oracle on the validation split earns +87.6 bps per trade, compared with +260 bps on the Tier-A test split. The lower oracle bps reflect the mechanics of an algorithmic collapse: during the Terra spiral, the route was impaired by unstable reference pricing, limiting achievable profit even with perfect foresight.

The cross-mechanism pattern supports the paper's central claim: price dislocations are common across mechanism types, but executable windows are rare in both cases. The oracle gap persists even when the mechanism differs, consistent with execution barriers being structural rather than event-specific.

The constraint is not whether stress occurs — it does, repeatedly and across diverse mechanisms — but whether route-provenance data supports Tier-A execution-labelled construction. Only Tier-A events provide that standard. Expanding Tier-A coverage to additional mechanism classes is the primary path to testing whether the oracle gap quantitatively generalises; Table 5 identifies the dominant execution friction that differs across class.

7 LIMITATIONS

Tier-A execution-labelled history is scarce. The test split covers one Tier-A execution-labelled stress event (SVB March 2023). Future work should extend Tier-A coverage to additional mechanism classes; the 18-event catalogue documents the data requirements. Models trained on fiat-reserve bank shocks may not generalise to algorithmic-reflexive or exchange-credit failure modes.

Historical catalogue is canonical, not exhaustive. The 18-event catalogue spans 2020–2023 and covers the major pre-2024 mechanism classes. It is designed as a benchmark taxonomy, not as a complete census of stablecoin disturbances through 2026. Post-2023 regulatory changes (e.g. the U.S. federal stablecoin framework), the emergence of synthetic-dollar and yield-bearing stablecoin products, and the changing role of Treasury/repo markets in backing stablecoin reserves are outside the empirical scope. Tier-B and Tier-C events support price and taxonomy claims only; exclusion of other events reflects the absence of route-complete depth data, not a judgment that those events were insignificant. The benchmark deliberately separates what can be claimed from Tier-A data from what can only be described at Tier-B or Tier-C.

Settlement and latency are proxied. CEX-to-CEX settlement latency is approximated via a fixed penalty parameter; true on-chain settlement delay and block confirmation times are not modelled. A more detailed treatment would require per-block latency data beyond the scope of this benchmark.

Venue-depth coverage. Binance USDM futures bookDepth for BTC-USDT is included in the public release for 2023 and 2024 benchmark windows, providing real L2 depth for the sell leg of the cross-quote route. The BTC-USDC buy leg and all 2022 windows use kline-synthesised depth (the 2022 Binance Vision archive does not include futures bookDepth). Full historical L2 for the Coinbase BTC-USD leg requires a Tardis subscription and is not committed to the public repository.

Model scope. The ladder covers tabular ML and statistics-based approaches. Deep sequence models, graph networks, and RL-based order-execution policies are outside scope and represent natural extensions. The oracle gap defines the challenge they must close.

8 CONCLUSION

Stablecoins promise price stability, but stress events create visible dislocations across venues. Those dislocations look like arbitrage on a chart. The central result of this paper is that they mostly are not: 34.3% of minutes in the USDC/SVB stress window show a cross-quote basis above 10 bps, but only 2.88% remain net-profitable after VWAP book-walking and costs at $10K notional — a 12× price-to-execution gap.

Profitable windows do exist: a hindsight oracle earns +162.2 bps per trade on the USDC-basis task, confirming that execution-labelled opportunities are not label artefacts under the documented route-provenance assumptions. On executable arbitrage tasks every active non-oracle strategy — price rules, ML classifiers, tree ensembles, and a direct expected-net-profit regressor — produces negative net bps out of sample; abstention remains the economic floor. The oracle gap exceeds 260 bps on executable arbitrage tasks and persists across fee regimes, settlement penalties, and notional sizes.

Stablecoin StressBench shows that the main challenge is not observing price dislocations, but predicting which dislocations survive execution. The benchmark defines an oracle gap between hindsight-executable opportunities and ex-ante deployable models. The historical catalogue shows why this benchmark matters beyond one event: stablecoin stress recurs across mechanisms, while only some episodes provide the depth required for execution-aware evaluation. Closing the oracle gap — through better microstructure features, improved calibration, or deeper sequence and network models — is the central challenge for future work on crypto market microstructure, financial ML validation, and execution-aware trading systems.

ACKNOWLEDGEMENTS

Anonymised for review.

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